Petfinder EDA Summary (Beginner-Friendly)

Goal: Understand how features relate to `AdoptionSpeed`, comparing different pet kinds (`Type`).

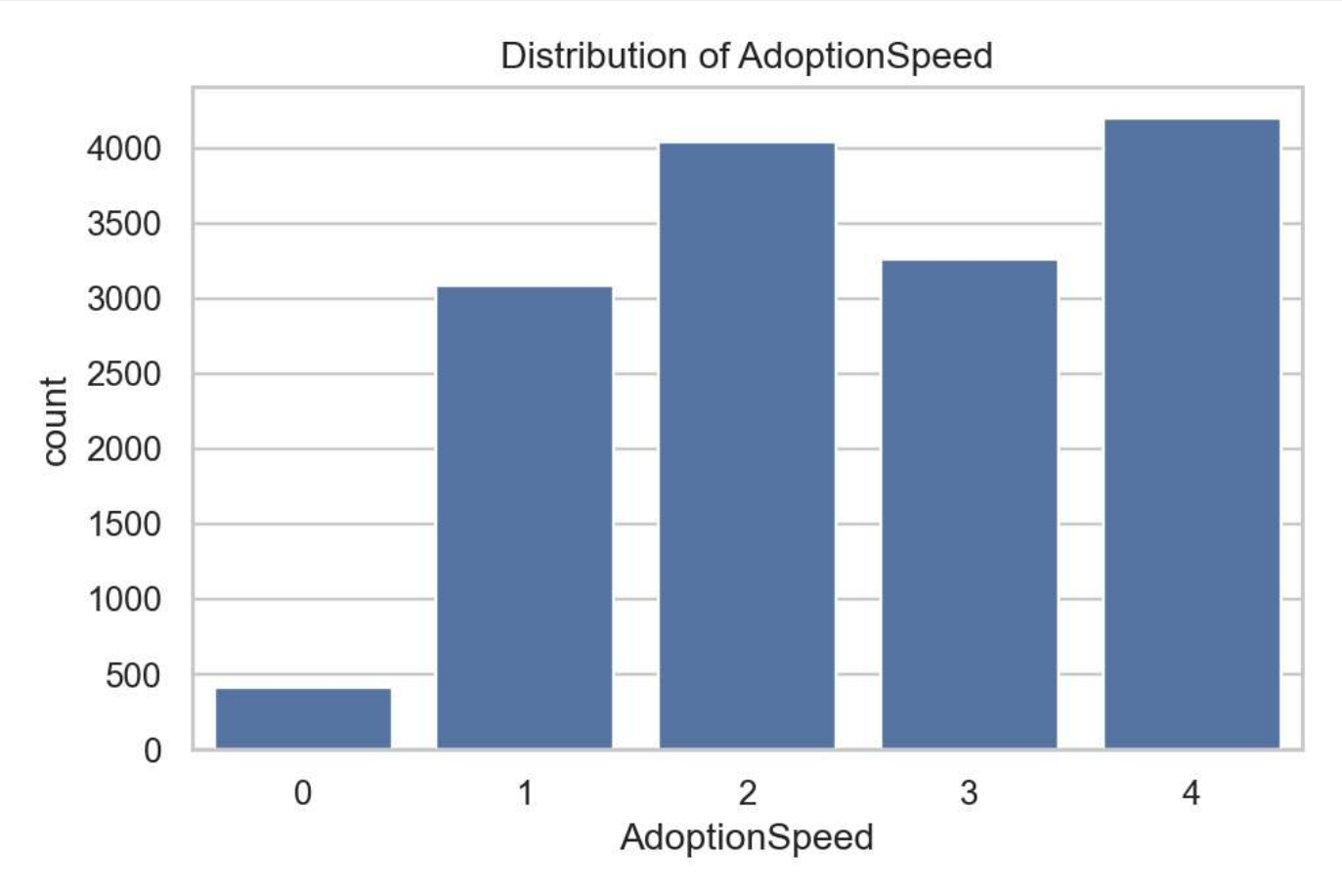
Data loaded from: C:\Users\paula\OneDrive\Maestria\Lab II\Carpeta VCS\Petfinder competicion\Competicion\Train\train.csv

Rows: 14993 **Columns:** 24

1) What is the target?

`AdoptionSpeed` is the adoption speed class (in the Petfinder competition it is usually an ordinal class).

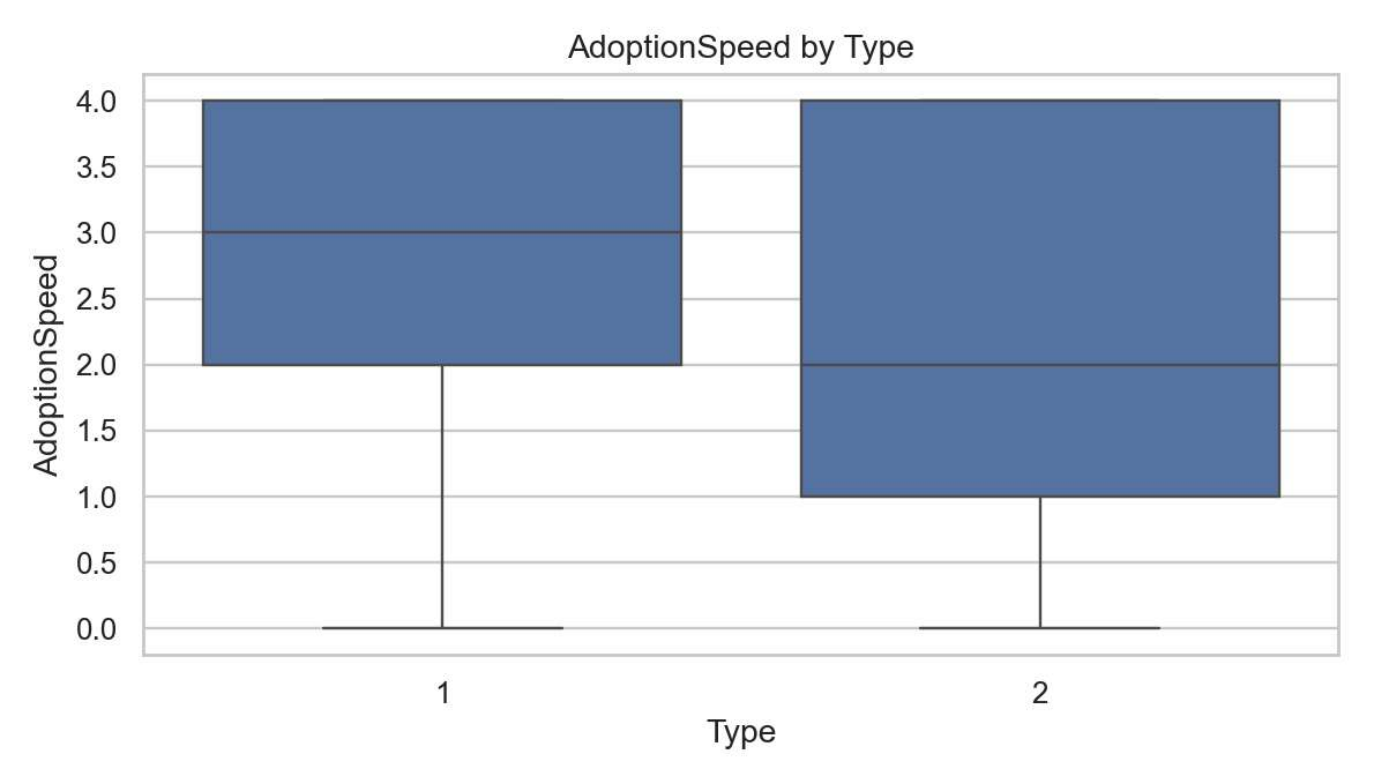
	count
AdoptionSpeed	
0	410
1	3090
2	4037
3	3259
4	4197

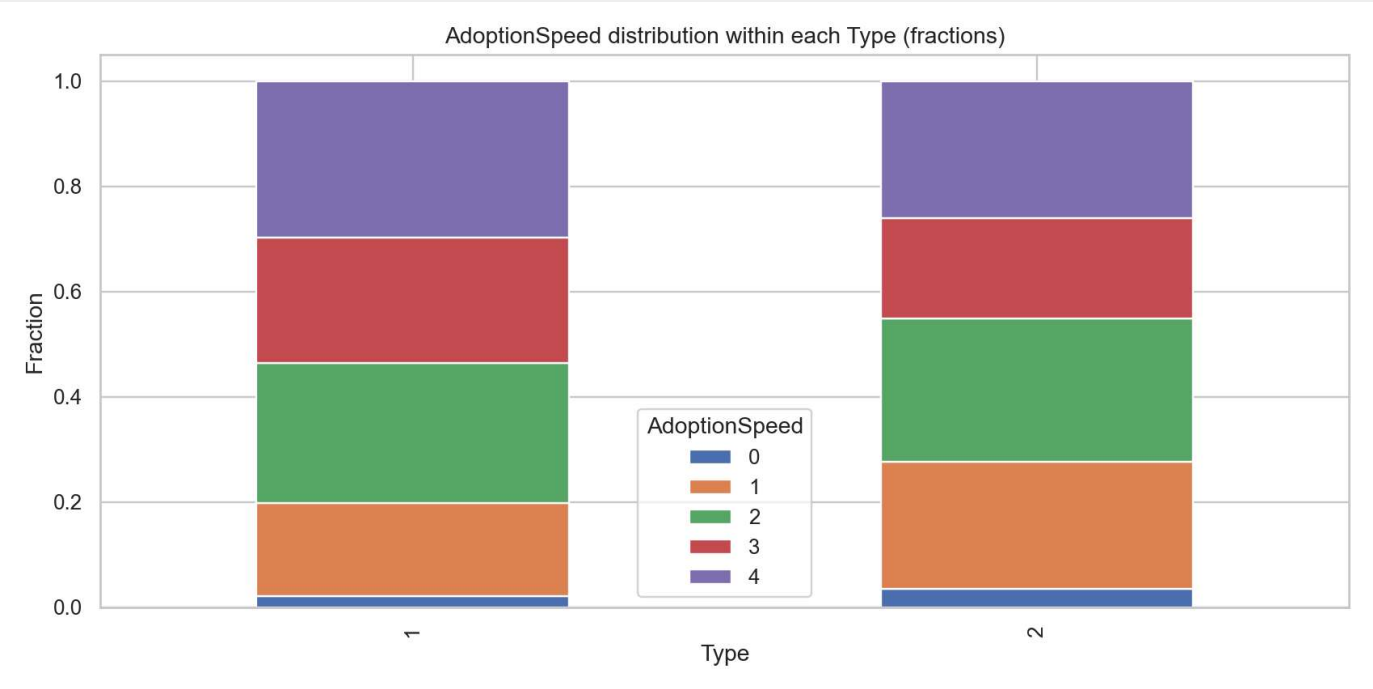


2) Pet kinds (dogs/cats/etc.)

This shows how many records you have per pet kind (`Type`).

	count
Type	
1	8132
2	6861



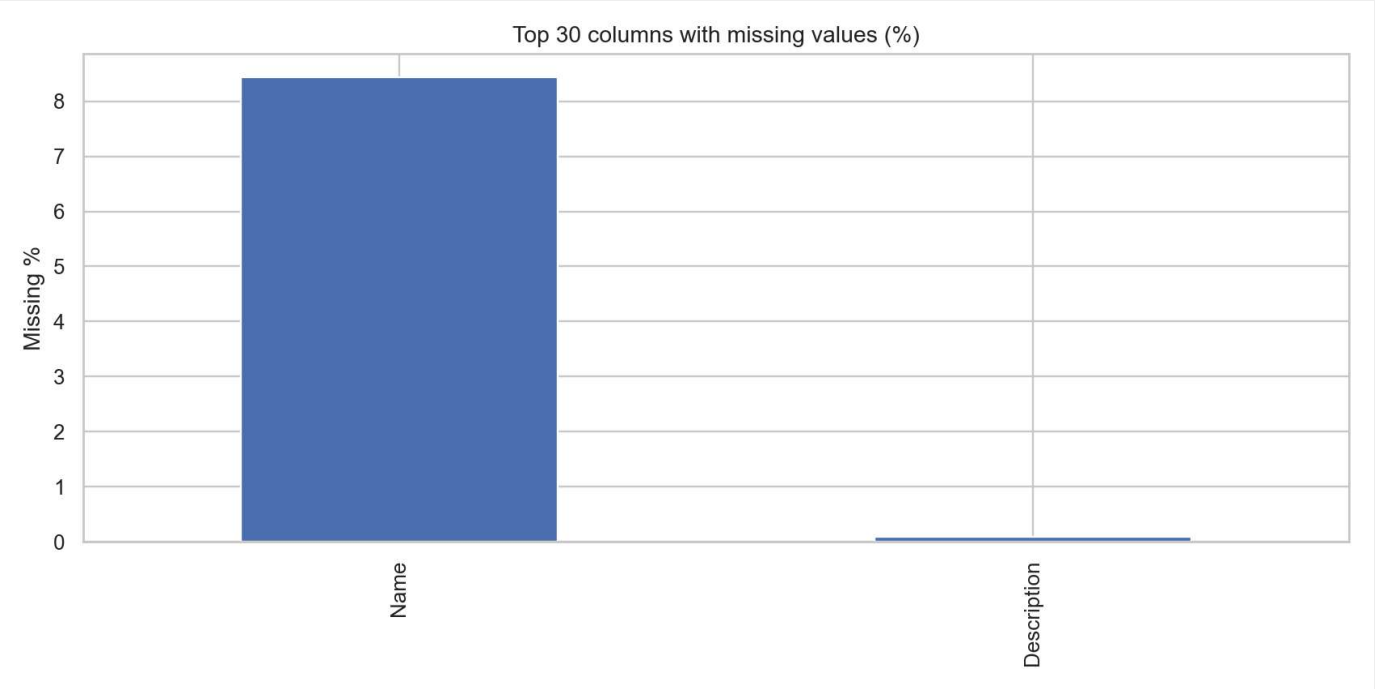


3) Data quality: missing values

Missing values can affect models and conclusions. Here are the columns with the most missing values.

	dtype	missing_%	n_unique
Name	object	8.44	9059
Description	object	0.09	14031

	dtype	missing_%	n_unique
PetID	object	0.00	14993
RescuerID	object	0.00	5595
Breed1	int64	0.00	176
Breed2	int64	0.00	135
Age	int64	0.00	106
Fee	int64	0.00	74
PhotoAmt	float64	0.00	31
Quantity	int64	0.00	19
State	int64	0.00	14
VideoAmt	int64	0.00	9
Color1	int64	0.00	7
Color2	int64	0.00	7
Color3	int64	0.00	6
AdoptionSpeed	int64	0.00	5
MaturitySize	int64	0.00	4
Gender	int64	0.00	3
FurLength	int64	0.00	3
Vaccinated	int64	0.00	3



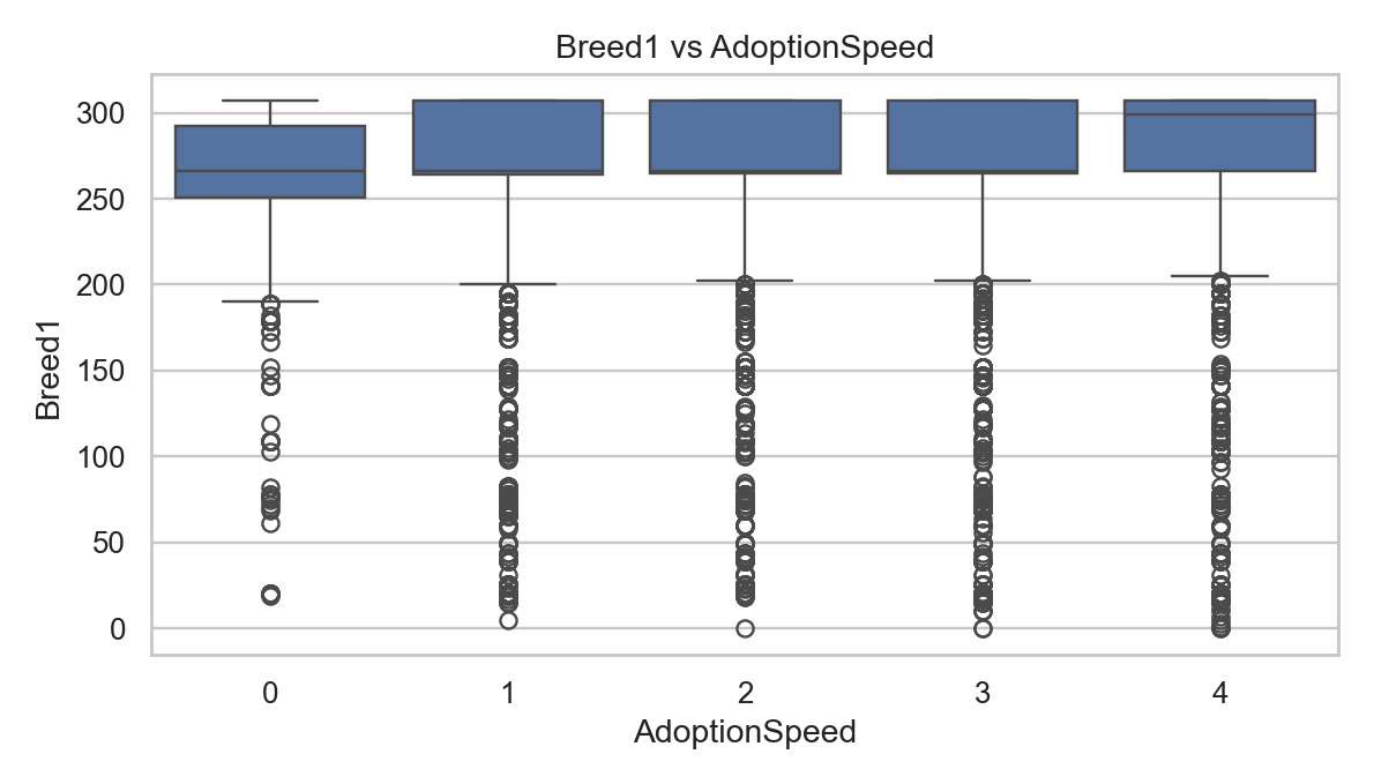
4) Which numeric features relate most to adoption speed?

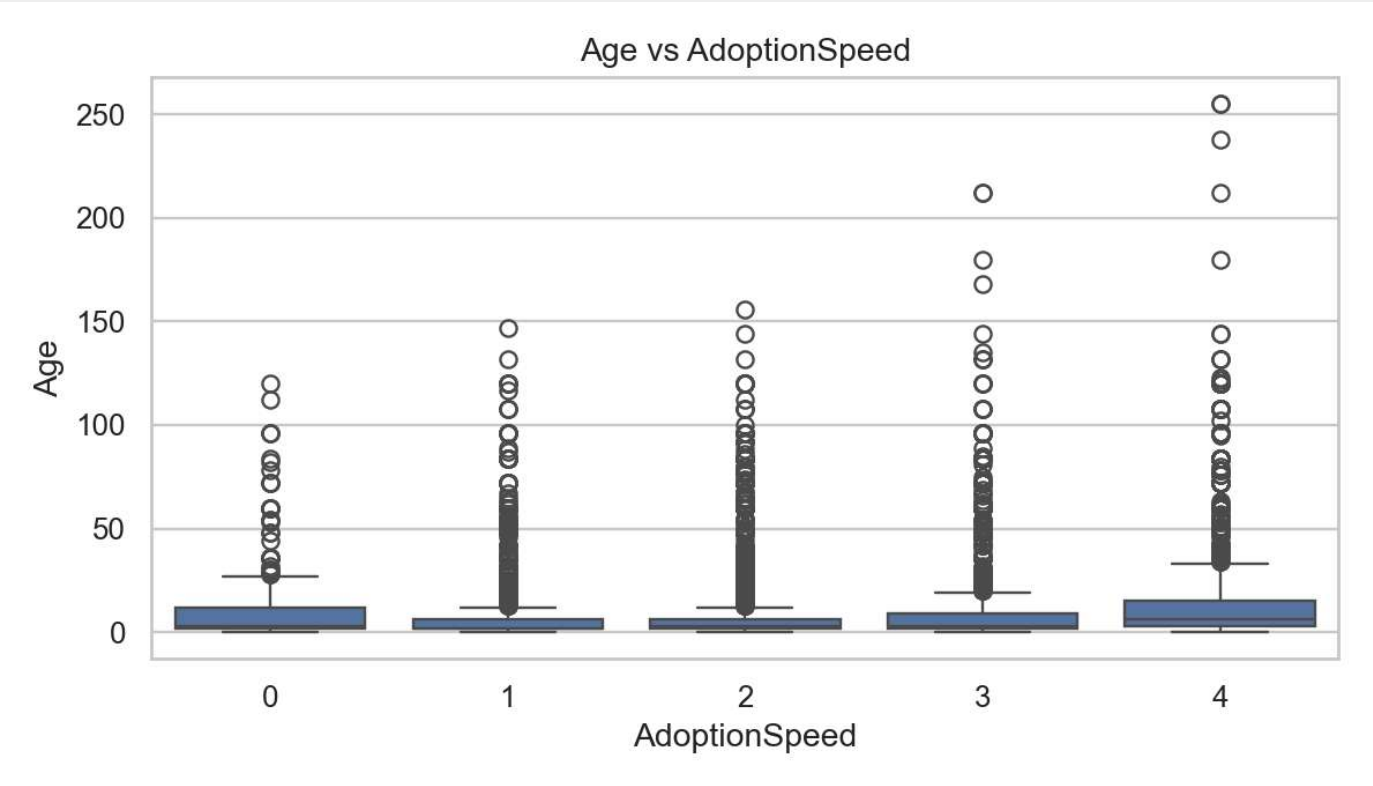
We compute a simple correlation between each numeric feature and `AdoptionSpeed`. This is a first screening (not final causation).

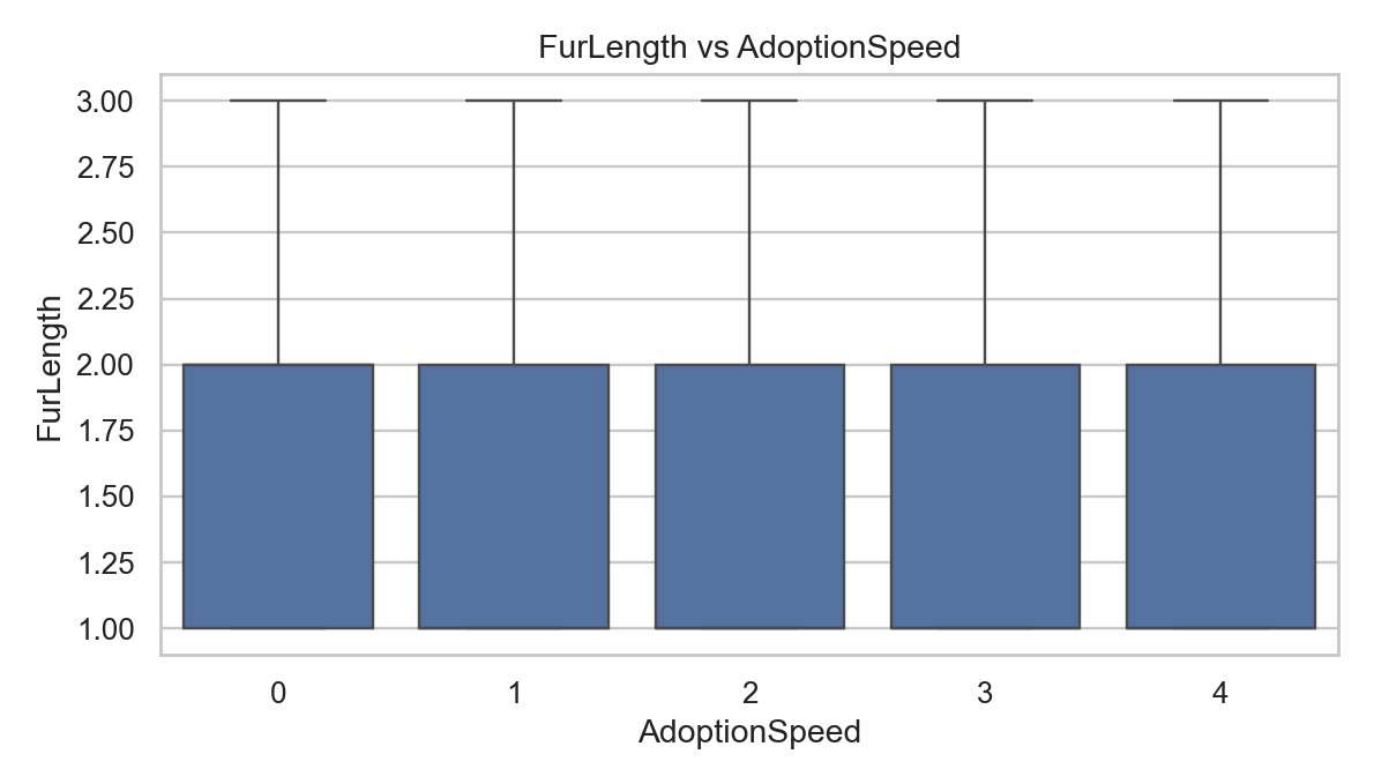
Top correlated numeric features:

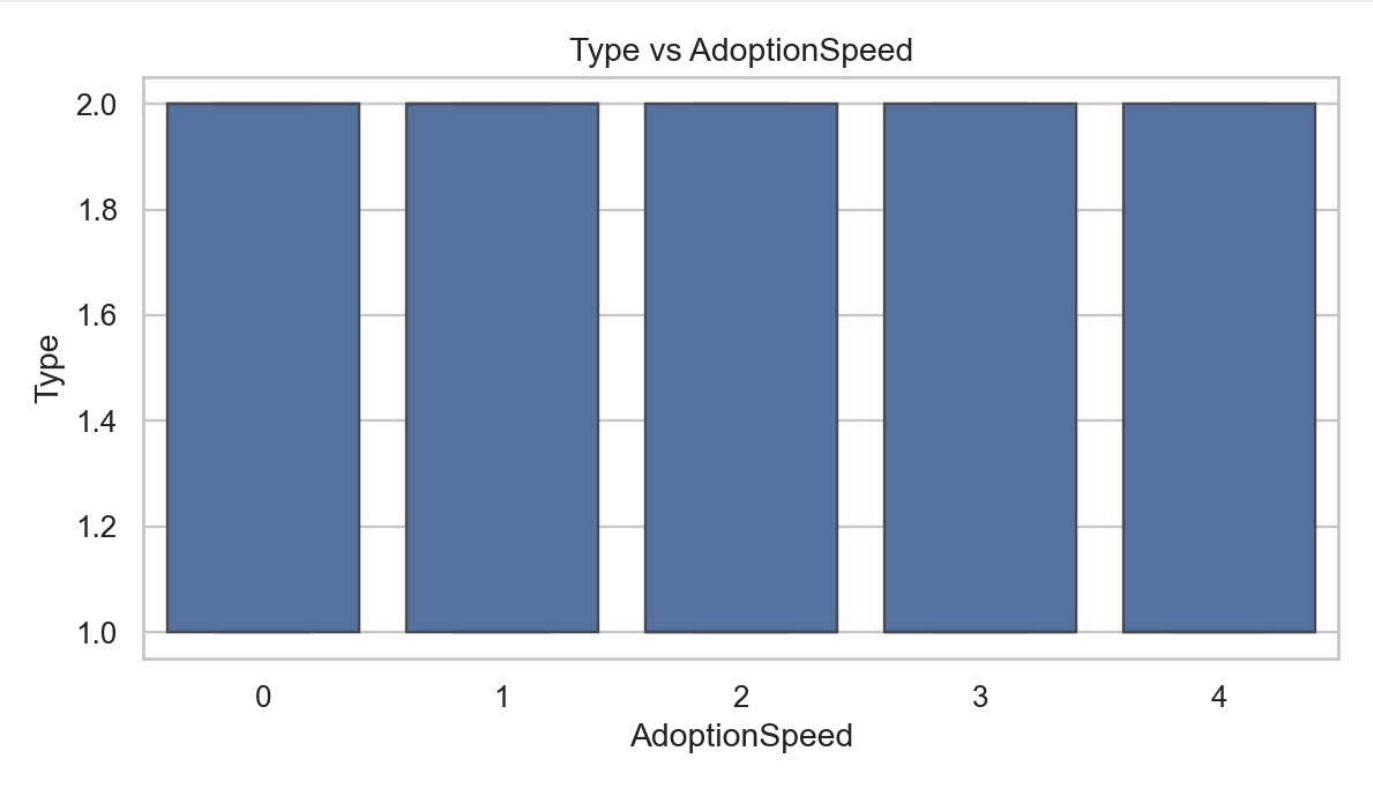
	corr_with_target
Breed1	0.107834
Age	0.100510
FurLength	-0.091337
Type	-0.091240
Sterilized	-0.083447
Quantity	0.062931
Vaccinated	-0.059043
Gender	0.057622
MaturitySize	0.045653
Color1	-0.044192
Color2	-0.038579
Health	0.029382

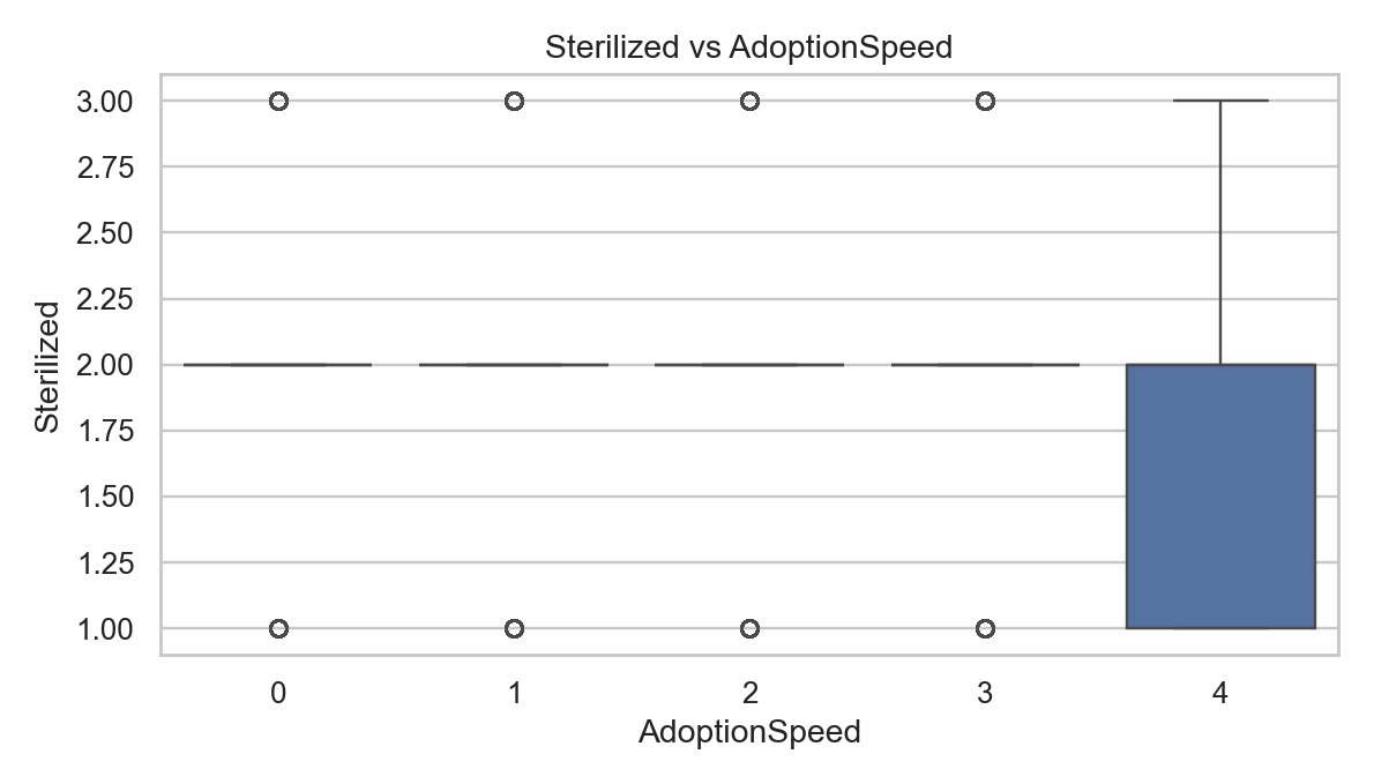
Plots for the top numeric features:

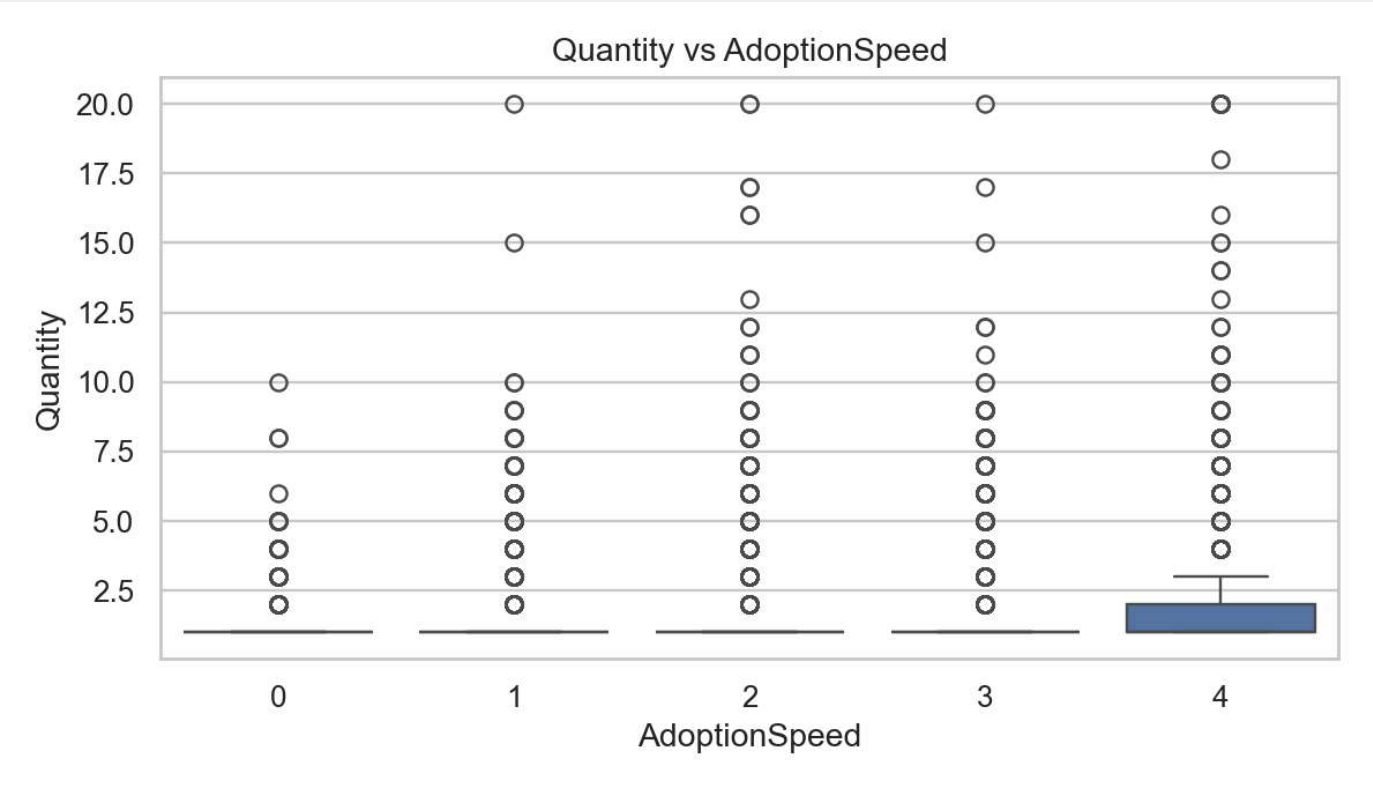

Breed1 vs AdoptionSpeed
Breed1
300
250
200
150
100
50
0
0
1
2
3
4
AdoptionSpeed


Age vs AdoptionSpeed
Age
250
200
150
100
50
0
0
1
2
3
4
AdoptionSpeed


FurLength vs AdoptionSpeed
FurLength
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00
0
1
2
3
4
AdoptionSpeed


Type vs AdoptionSpeed
Type
2.0
1.8
1.6
1.4
1.2
1.0
0
1
2
3
4
AdoptionSpeed





5) Comprehensive report (automatic)

A full automatic EDA report was created as: `petfinder_profile_report.html`.